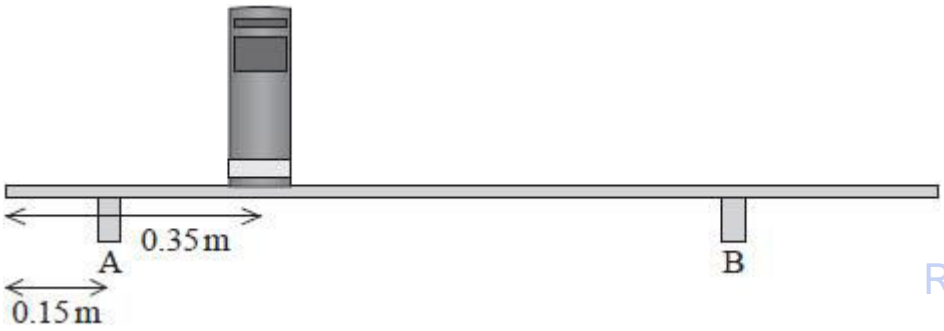
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Questions

Q1.

Two brackets, A and B, support a shelf of length 1.2 m. Bracket A is positioned 0.15 m from the left-hand end of the shelf. A book is placed 0.35 m from the left-hand end of the shelf as shown.



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(a) The normal contact forces of each bracket on the shelf are equal.

Determine the distance of bracket B from the left-hand end of the shelf.
weight of book = 8.5 N
weight of shelf = 14 N

(5)

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Distance = ...

(b) Bracket B is moved closer to the left-hand end of the shelf.

Explain the effect on the magnitude of the normal contact force of bracket B on the shelf.

(2)

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(Total for question = 7 marks)

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Q2.

All quantities may be expressed in terms of SI base units.

Which of the following are the base units for the moment of a force?

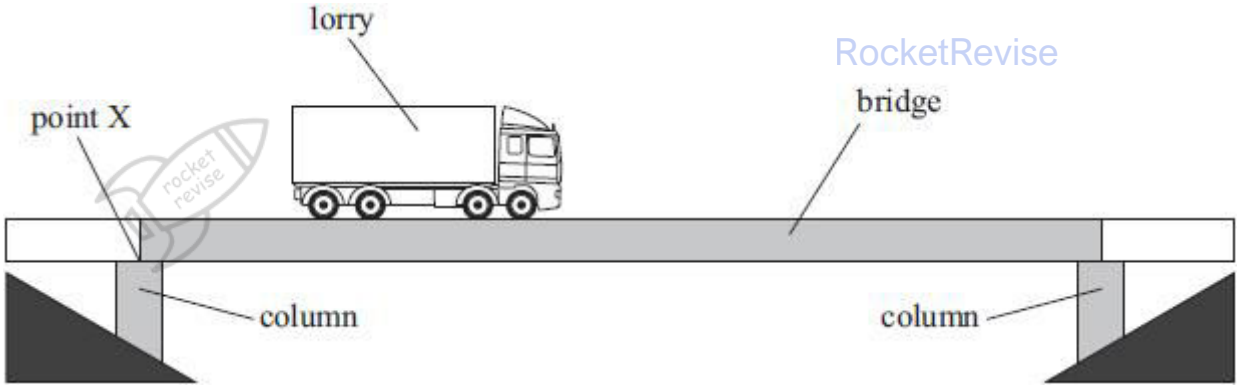
(1)

- ☒ **A** kg m s^{-2}
- ☒ **B** kg m^2 s^{-2}
- ☒ **C** kg m s^{-1}
- ☒ **D** kg m^2 s^{-3}

(Total for question = 1 mark)

Q3.

A lorry is crossing a bridge. The bridge is supported by two columns, as shown.

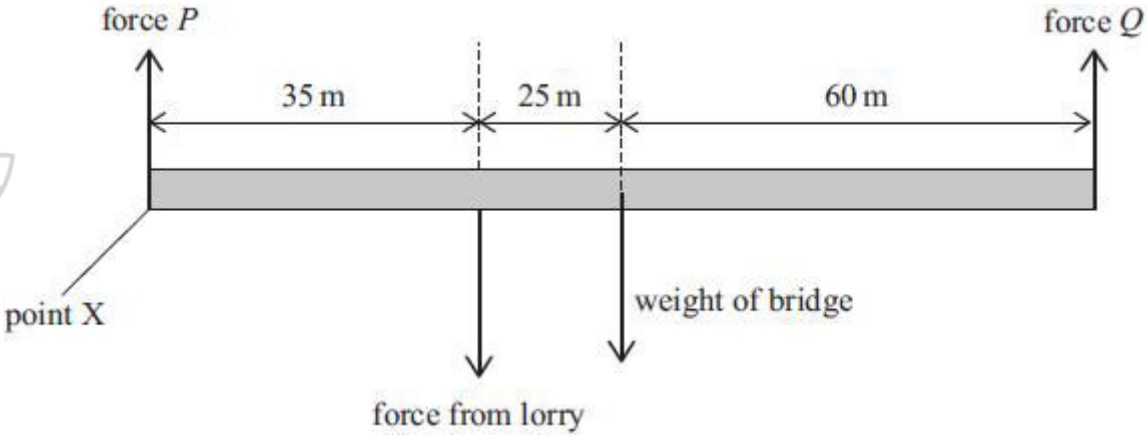


Forces *P* and *Q* act on the bridge from the columns. The centre of gravity of the bridge is at its centre. At a particular time the lorry is 35 m from point X.

A simplified diagram showing the positions of the forces on the bridge is shown below.

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(a) Calculate the magnitudes of the forces *P* and *Q*.

You should take moments about point X.
force from lorry = 4.2×10^5 N
weight of bridge = 9.8×10^5 N

(5)

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P = ...

Q = ...

(b) Explain why force *Q* increases as the lorry moves across the bridge.

(5)

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(Total for question = 10 marks)

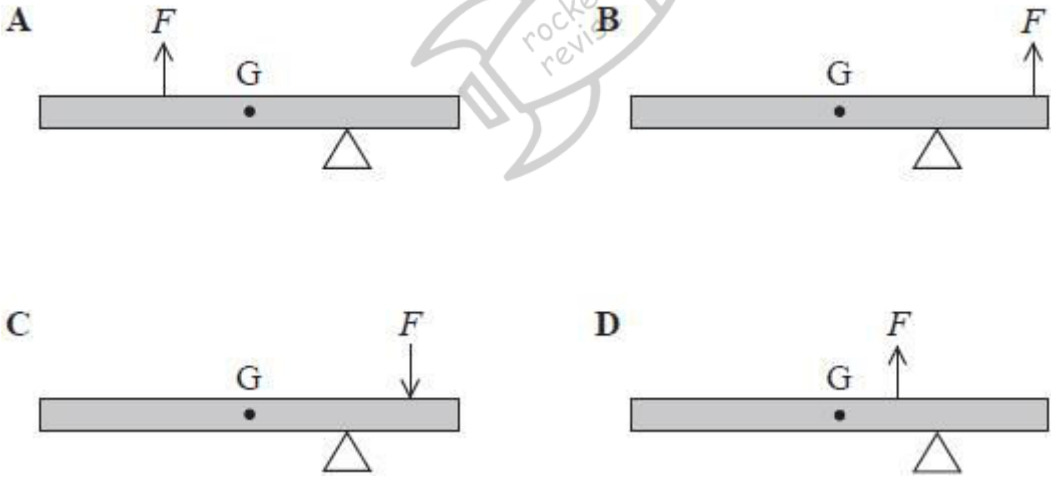
Q4.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A beam is supported by a pivot as shown in the diagrams. The centre of gravity of the beam is at G. The beam is acted on by a force *F*.

Which diagram shows an arrangement where the beam could **not** be in equilibrium?



- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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(Total for question = 1 mark)

Q5.

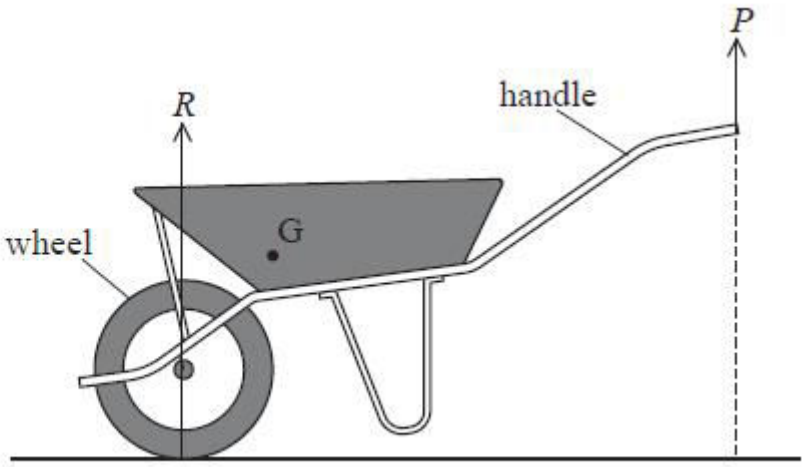
A woman supports a stationary wheelbarrow by exerting an upward force *P* on the handles. An upward force *R* from the ground acts on the wheel, as shown.

The centre of gravity G of the wheelbarrow is marked on the diagram.

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(Source: © Cavan Images/Alamy Stock Photo)

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(a) Explain, by considering moments about G, why *P* is less than *R*.

(3)

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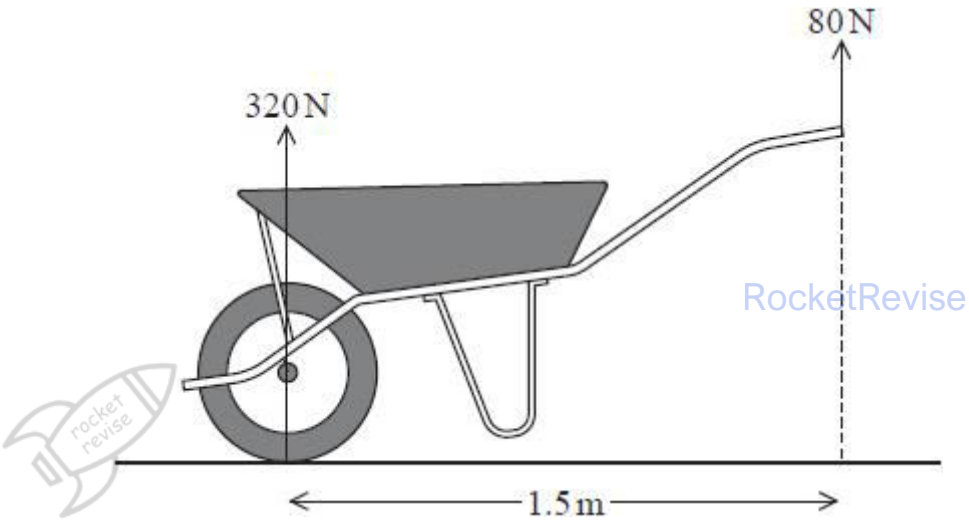
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(b) The horizontal distance between the centre of the wheel and the handles is 1.5 m.

The magnitudes of *P* and *R* are as shown.



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Determine the horizontal distance between the centre of gravity of the wheelbarrow and the centre of the wheel.

(4)

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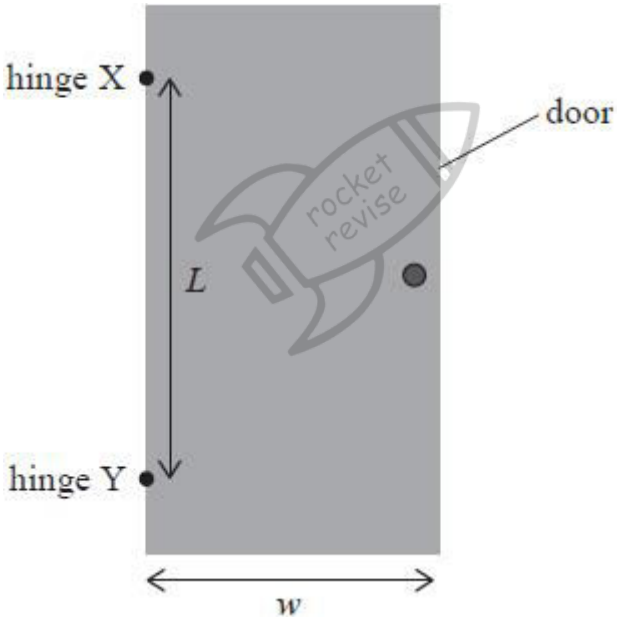
Horizontal distance = ...

(Total for question = 7 marks)

Q6.

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A uniform door of width *w* is supported by two hinges, X and Y.
X and Y are a vertical distance *L* apart as shown.



(a) State the conditions needed for the door to be in equilibrium.

(2)

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(b) Give the meaning of the term centre of gravity.

(1)

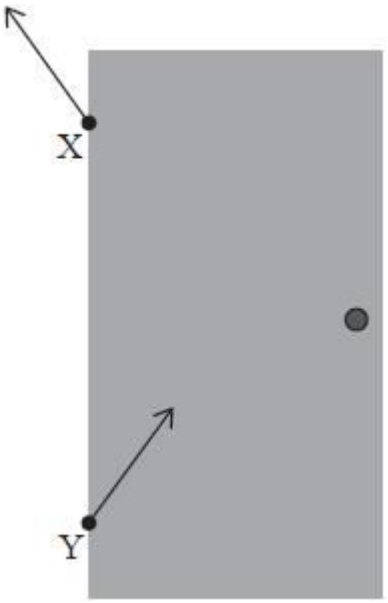
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(c) The diagram shows the force acting on the door at each hinge.

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Not to scale

RocketRevise

The mass of the door is 14.4 kg.
The weight of the door causes a moment about each hinge.

(i) Show that the horizontal component of the force of hinge Y on the door is about 40 N.

L = 1.60 m
w = 0.85 m

(5)

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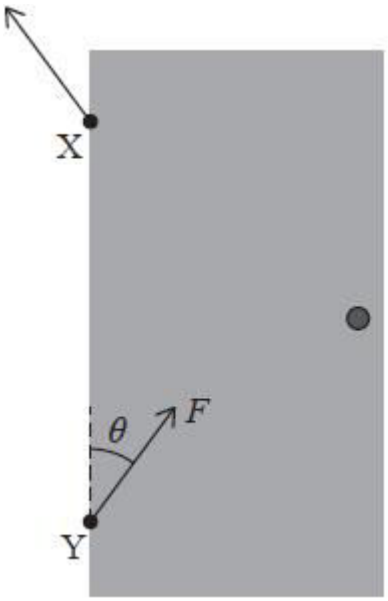
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(ii) The vertical component of the force of each hinge on the door is equal to half the weight of the door.

Hinge Y exerts a force F on the door at an angle θ to the vertical, as shown.

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Not to scale

RocketRevise

Determine the force F and the angle θ.

(4)

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F = ...

θ = ...

(iii) An identical door is supported by hinges that are much closer together.
Explain how a smaller value of L affects F and θ.
You should consider moments about point X.

(4)

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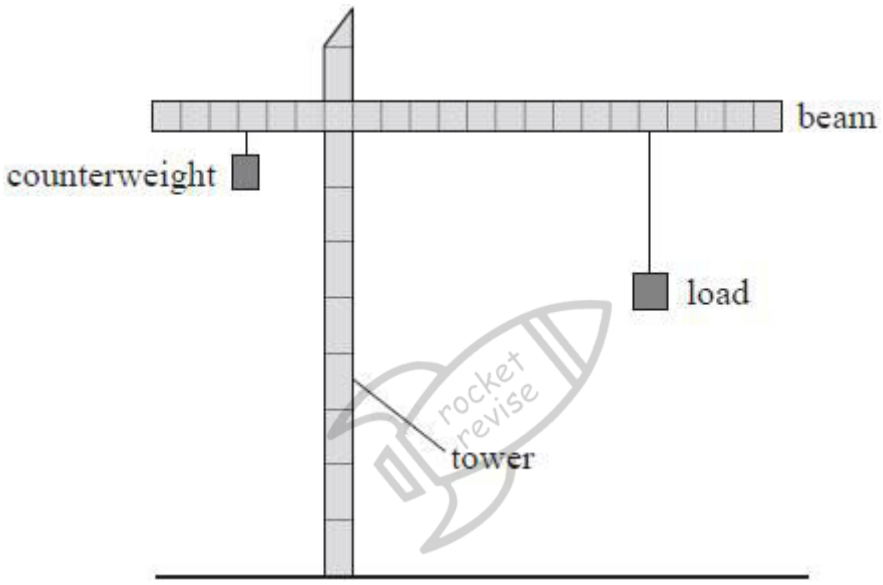
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(Total for question = 16 marks)

Q7.

The diagram shows a crane lifting a load. There is a large mass called a counterweight attached to the crane, on the opposite side of the tower to the load.



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(a) The crane has an electric motor with an efficiency of 47%.

The crane lifts a load of 4.4×10^4 N. The load moves through a vertical distance of 15 m in a time of 70 seconds.
Determine the average power input to the electric motor.

(4)

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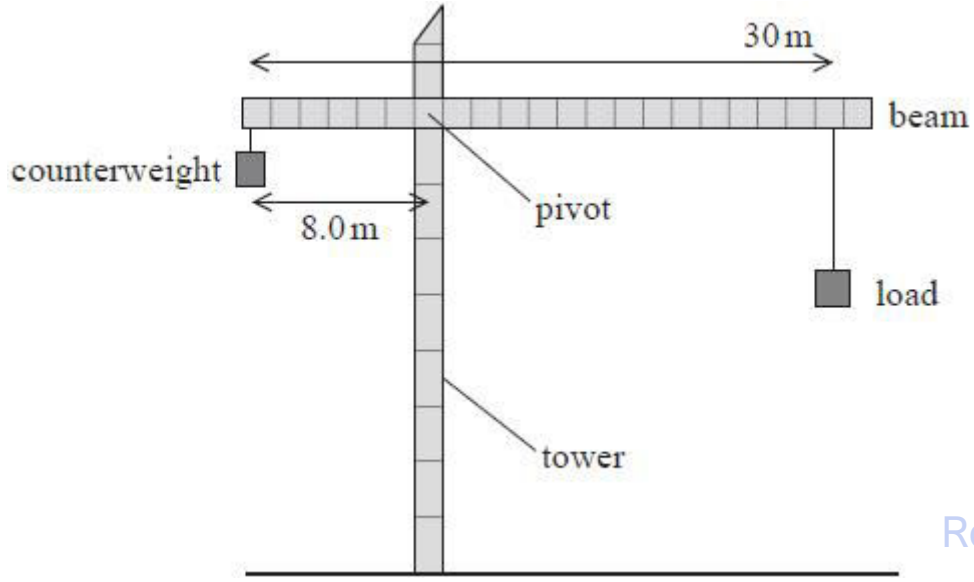
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Average power input = ...

(b) The horizontal distance from the load to the tower can be changed. The horizontal distance from the counterweight to the tower can also be changed.

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(i) Explain how the counterweight needs to move when the load is moved away from the tower.

(4)

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(ii) The beam is uniform with a weight of 3.0×10^4 N.
The counterweight has a weight of 1.1×10^5 N.
The load has a weight of 4.4×10^4 N.
Deduce whether the load can be moved to the end of the beam without the crane toppling.

(5)

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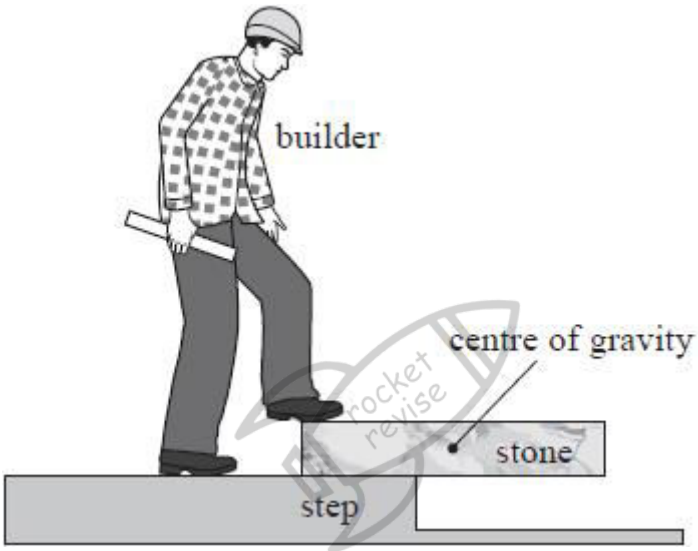
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(Total for question = 13 marks)

Q8.

A builder is making a path using uniform rectangular stones. A stone is resting horizontally on a step and is held in equilibrium by the builder's foot.

The centre of gravity of the stone is at its centre, as shown.



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(a) (i) State what is meant by equilibrium.

(2)

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(ii) State what is meant by the centre of gravity of an object.

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(1)

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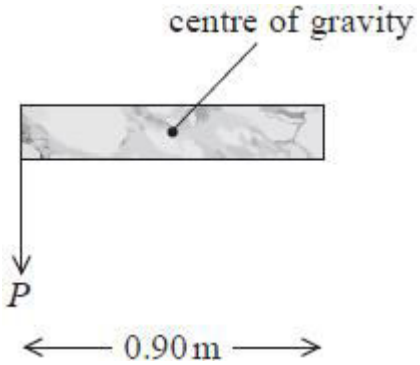
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(b) The builder applies the minimum force *P* that will keep the stone in equilibrium.

(i) The position and direction of *P* are shown on the diagram below.

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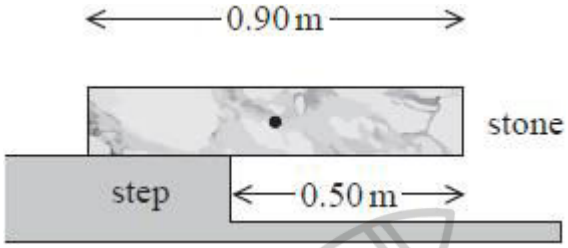


Complete the diagram above by adding labelled arrows to show the positions and directions of the other forces acting on the stone.

(2)

(ii) The stone has a weight of 415 N and a length of 0.90 m.
The length of the stone beyond the edge of the step is 0.50 m, as shown below.

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The centre of gravity is at the centre of the stone.
Calculate the magnitude of *P*.

(3)

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Magnitude of *P* = ...

(Total for question = 8 marks)

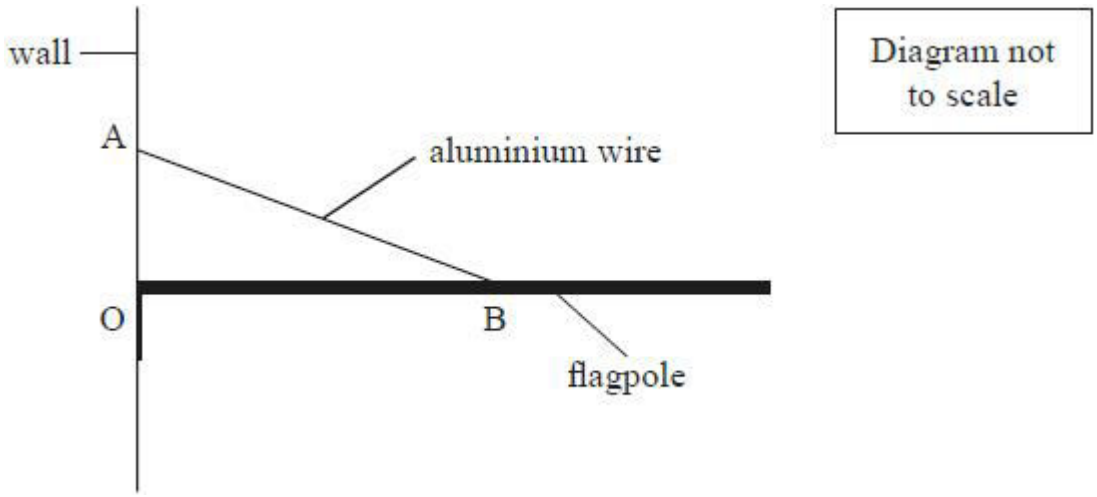
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Q9.

A uniform, horizontal flagpole is connected by a hinge to a wall at position O. An aluminium wire connects the pole to the wall at A, as shown.

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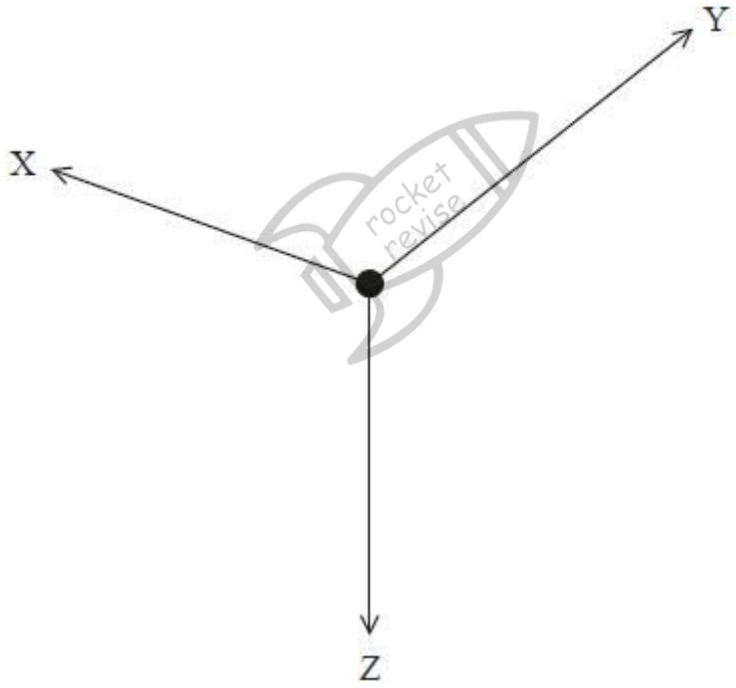
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(a) A free-body force diagram for the flagpole is shown below.

Identify the forces X, Y and Z.

(3)



X .....

Y .....

Z .....

(b) The aluminium wire will break if the tension in the wire exceeds 350 N.

The wire is attached to the flagpole at B, 0.8 m from the wall.
The wire is at an angle of 20° to the flagpole.
Assess whether the wire will break. You should use the principle of moments, taking moments about O.
length of flagpole = 1.2 m
mass of flagpole and flag = 15 kg

(3)

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(Total for question = 6 marks)

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Q10.

A coat hook is attached to a smooth door by a screw as shown.

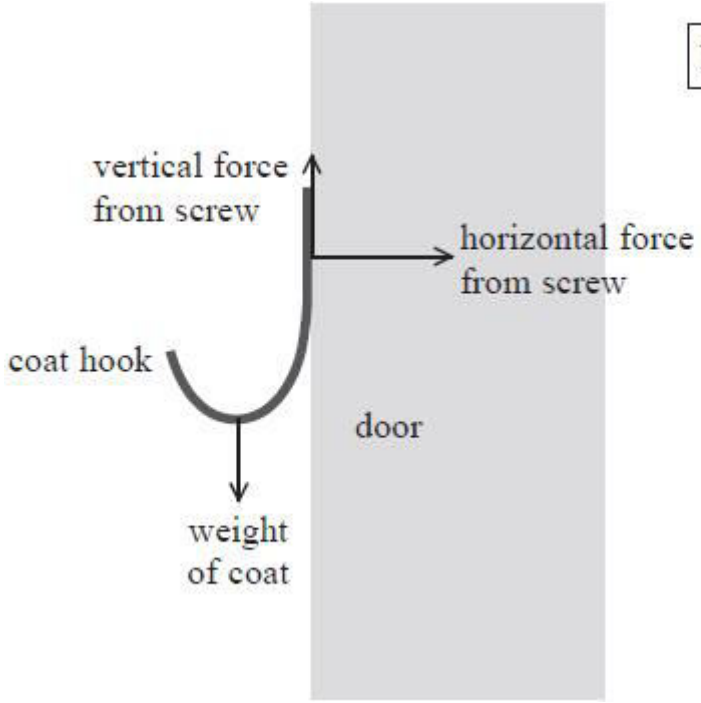


(a) The diagram below shows three of the forces that act on the coat hook when a coat is hung from it. The weight of the coat hook may be neglected.

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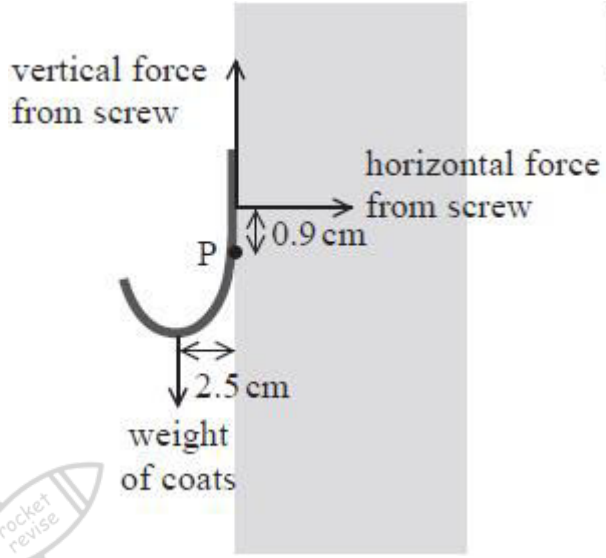


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Add a labelled arrow to the diagram to show the additional force required for the coat hook to be in equilibrium.

(2)

(b) If too many coats are hung on the coat hook, the hook will rotate and pull the screw out of the door. Point P is the position of the pivot as shown.



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The maximum horizontal force from the screw is 150 N.

The mass of one coat is 2.6 kg.

Deduce whether a person could hang more than two of these coats from the hook.

(4)

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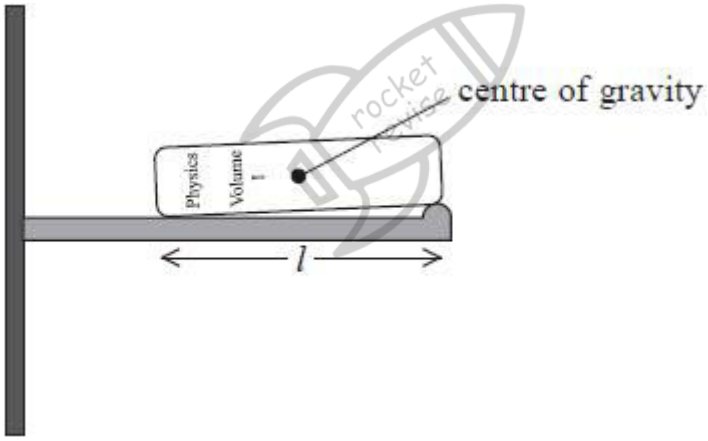
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(Total for question = 6 marks)

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Q11.

A book of length *l* and weight *W* is positioned on a shelf as shown.



(a) State what is meant by centre of gravity.

(1)

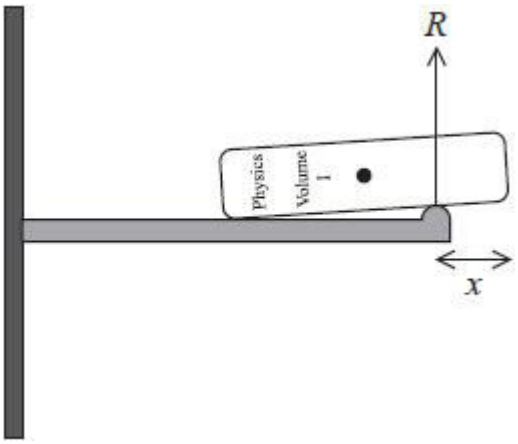
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(b) There are contact forces at the two positions where the book touches the shelf.

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The book is moved a distance *x* to the right as shown.



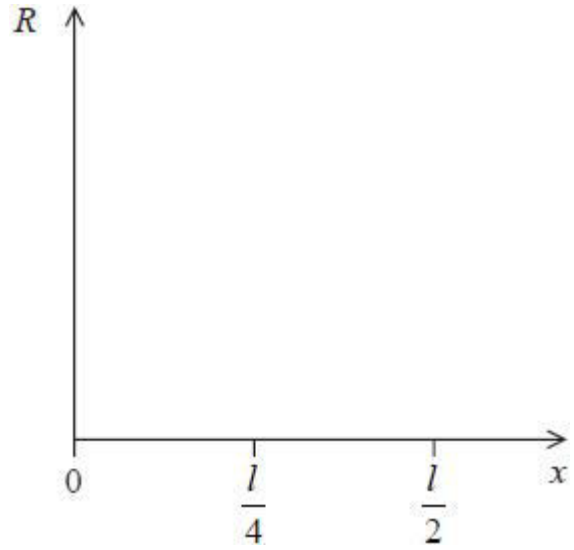
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The magnitude of the contact force R varies with x.

Sketch on the axes below the variation of R with x between 0 and $\frac{l}{2}$. You should mark the maximum value of R onto the axes.

(3)



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(c) Explain why the book will not remain on the shelf if $x > \frac{l}{2}$

(2)

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(Total for question = 6 marks)

Q12.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of the following describes the centre of gravity of an object?

- ☐ **A** the point where the mass of the object is located
- ☐ **B** the point where the mass of the object is concentrated
- ☐ **C** the point where the weight of the object is located

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☒ **D** the point where the weight of the object may be considered to act

(Total for question = 1 mark)

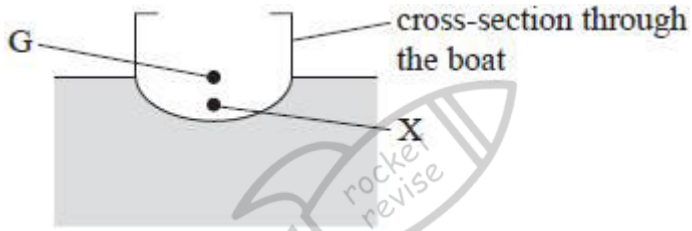
Q13.

For a boat to maintain a constant upright position in the water, the weight *W* and upthrust *U* should have the same line of action.

W acts through G, the centre of gravity of the boat.

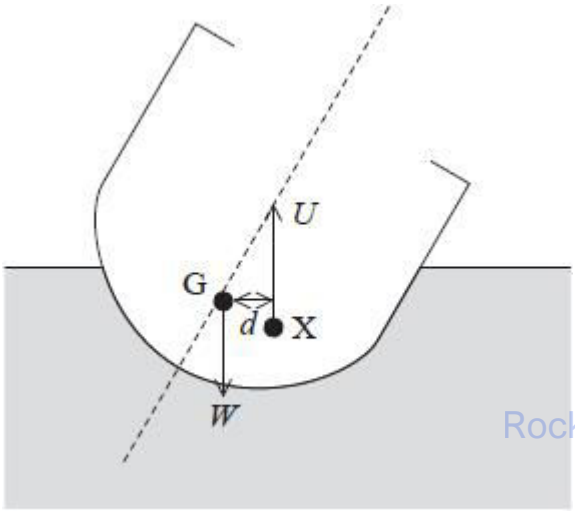
U acts through X, the centre of gravity of the volume of water displaced by the boat.

G and X are shown on the diagram.



(a) A sudden gust of wind applies a force to the side of the boat, causing it to tilt.

The lines of action of *W* and *U* move apart a distance *d*, as shown below.



(i) Explain the effect of the moment *Ud*.

(2)

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(ii) Explain what would happen to the boat in windy weather if the centre of gravity G were higher.

(3)

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(b) Some boats use a ballast tank, which is a refillable tank of water in the base of the boat, to improve stability.



Ballast tank empty Ballast tank full

RocketRevise

(i) Explain why the position of X for the ship when it has a full ballast tank is lower than when the ship has an empty ballast tank.

(4)

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(ii) Explain a disadvantage of using a full ballast tank when the boat is moving through the water.

(2)

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(Total for question = 11 marks)

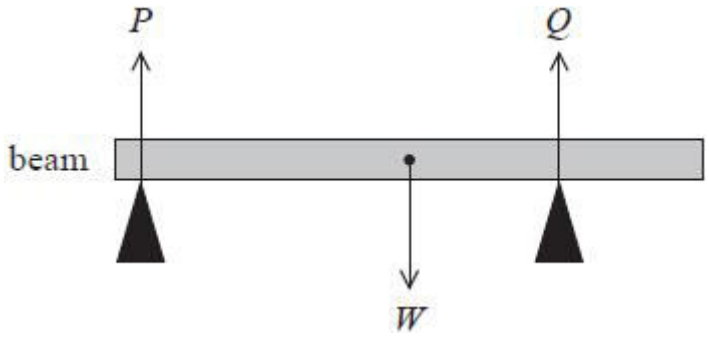
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Q14.

Select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A beam is balanced on two supports as shown.



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The beam has a weight W and the reaction forces at the two supports are P and Q.

Which of the following statements about the magnitudes of the forces is correct?

- ☒ **A** $P > Q$
- ☒ **B** $Q > W$
- ☒ **C** $Q > P$
- ☒ **D** $(P + Q) > W$

(Total for question = 1 mark)

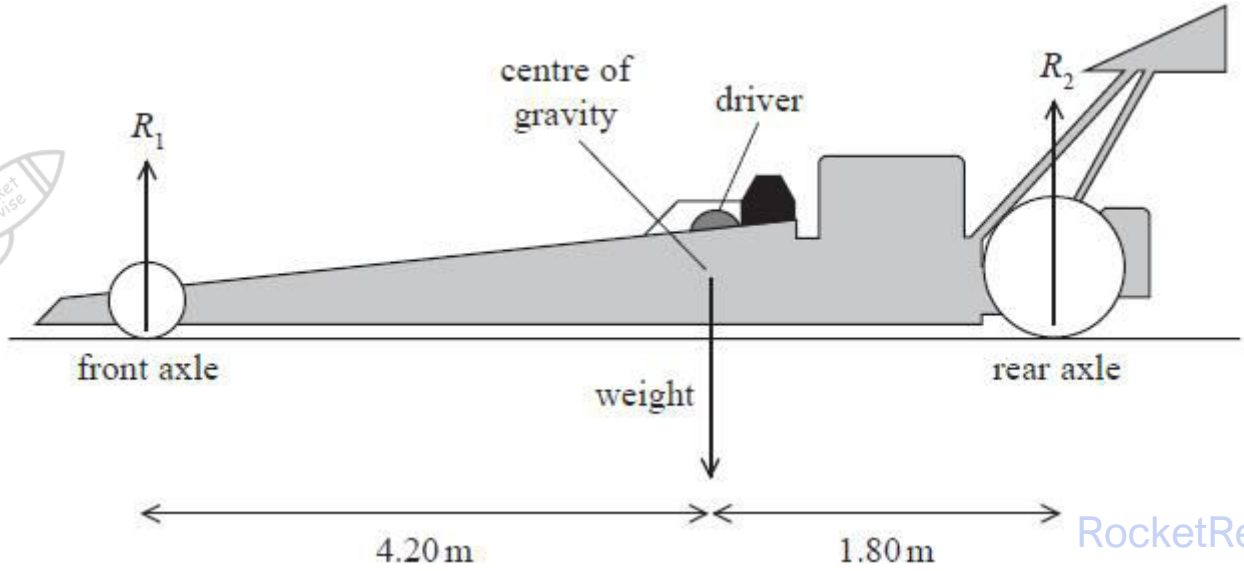
Q15.

A dragster is a racing car designed for a very short race along a completely straight track, so must be able to accelerate at a very high rate. The dragster and driver shown below have a combined weight of 1.23×10^4 N. The centre of gravity is 1.80 m in front of the rear axle.

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The front axle of the dragster is 6.00 m from the rear axle.

(a) Calculate the reaction forces R_1 and R_2, shown on the diagram, when the dragster is stationary and not accelerating.

(3)

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R_1 = ...

R_2 = ...

(b) When the dragster starts, there is a driving force that gives the dragster an initial forward acceleration of 5.50 g.

Calculate the initial driving force on the dragster.

(2)

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Initial driving force = ...

(c) The power from the car's engine is constant.

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Explain how the force from the engine varies as the car accelerates.

(2)

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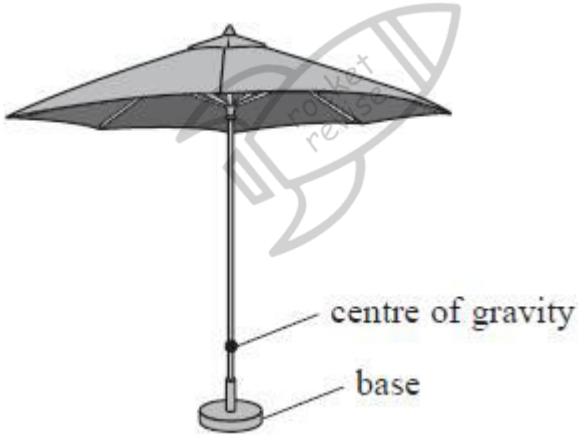
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(Total for question = 7 marks)

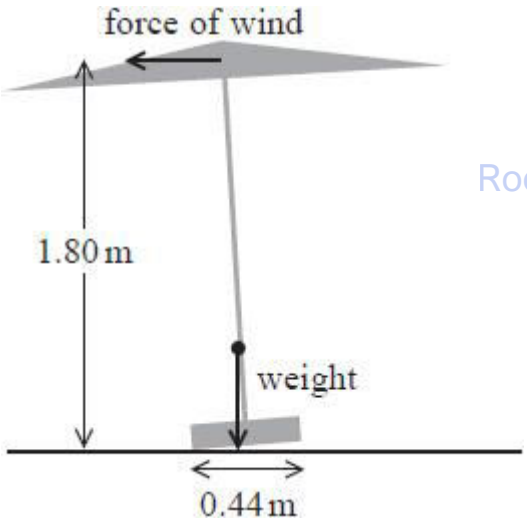
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Q16.

A large parasol has been set up on a windy day. The centre of gravity of the parasol is vertically above the centre of the base. The bottom of the parasol starts to lift from the ground as shown. The weight of the parasol is 110 N.



(a) The force of the wind is 14 N in a horizontal direction.



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Explain why the parasol will topple. Your answer should include a calculation.

(4)

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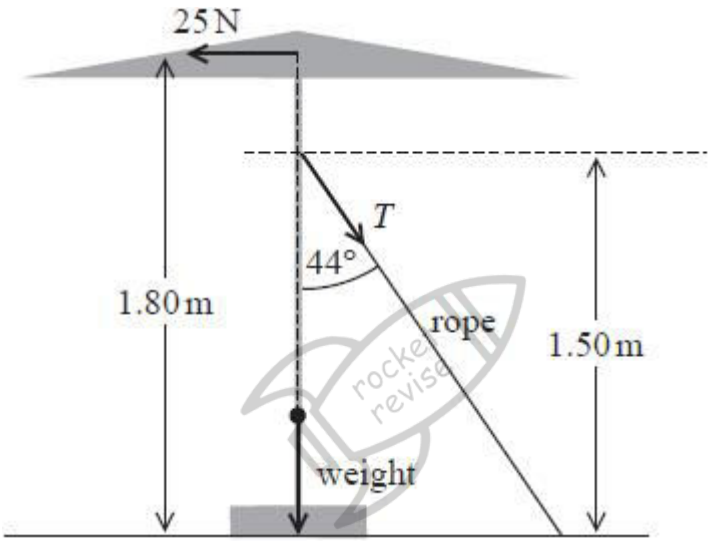
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(b) To prevent the parasol from toppling, a rope is attached to the parasol at 1.50 m from the ground as shown. The rope makes an angle of 44° to the vertical.



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The horizontal force from the wind is now 25 N.
Determine, by taking moments about the centre of the base, the vertical force that the base exerts on the ground.
Assume that the force which the ground exerts on the base acts through the midpoint of the base.

(5)

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Force exerted on the ground = ...

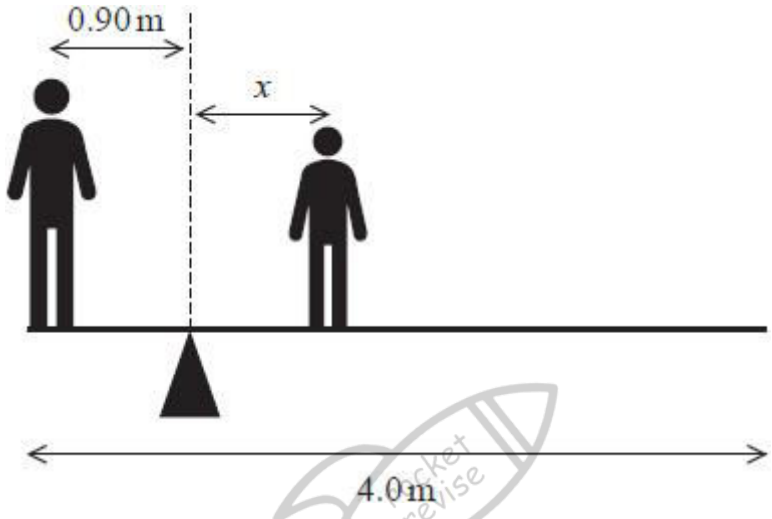
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(Total for question = 9 marks)

Q17.

A uniform plank of length 4.0 m is pivoted 0.90 m from one end. The weight of the plank is 250 N. A person of weight 950 N stands at one end of the plank. A person of weight 650 N stands a distance *x* from the pivot so that the plank is in equilibrium, as shown.



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(a) Add to the diagram to show the forces acting on the plank.

(2)

(b) Calculate the distance *x*.

(3)

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x = ...

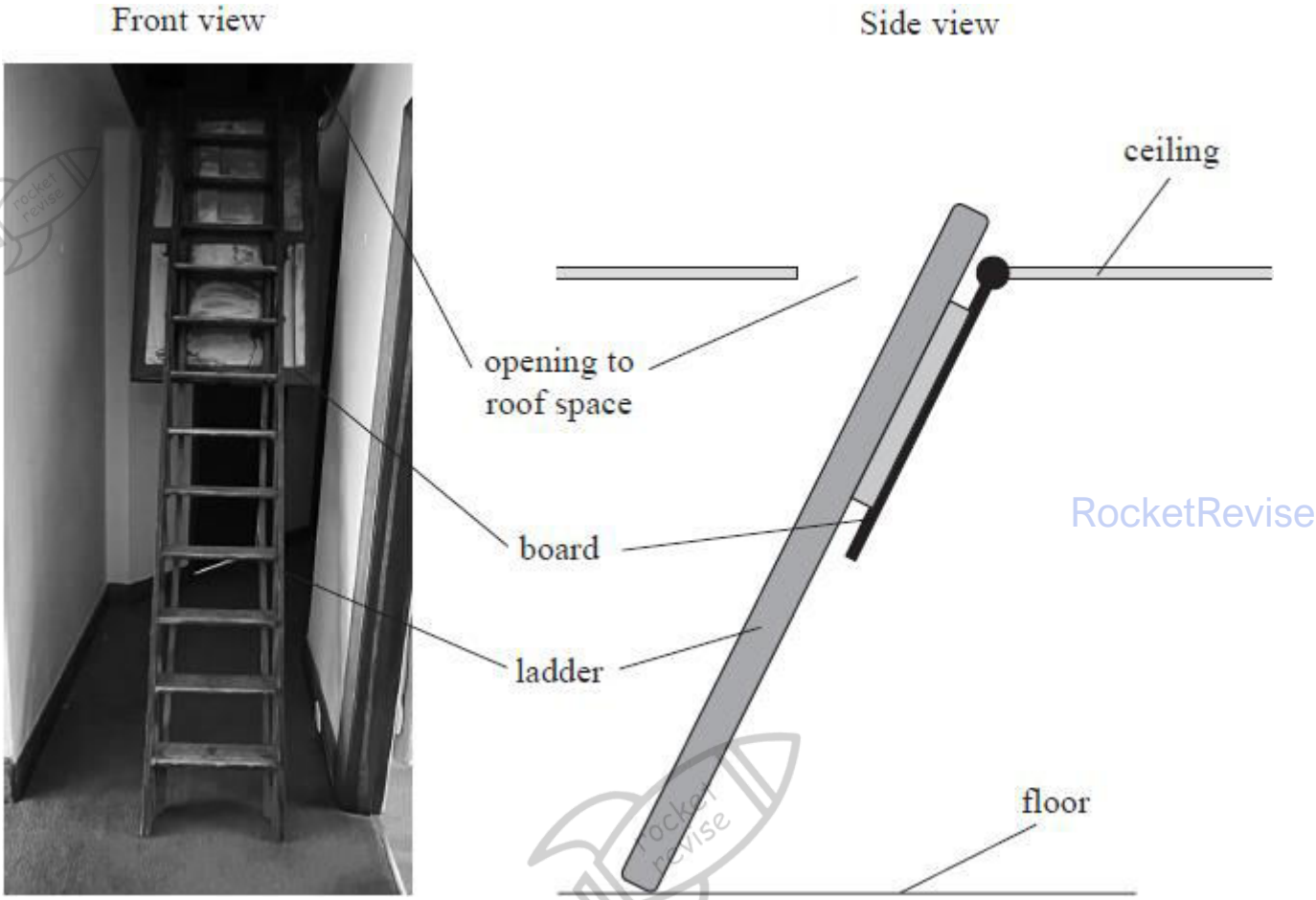
(Total for question = 5 marks)

Q18.

The photograph shows a ladder that a person can use to climb into the roof space of a building.

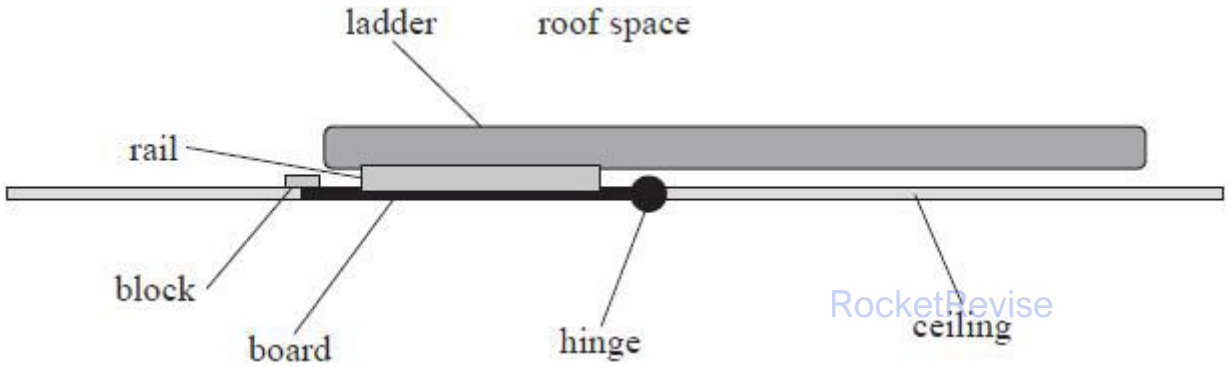
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When not in use, the ladder is stored in the roof space and the opening to the roof space is covered by a board. The board is hinged to the ceiling at one end.

The ladder is attached to the board by a rail, as shown below. A small block prevents the board from rotating into the roof space.



(a) State what is meant by the centre of gravity of an object.

(1)

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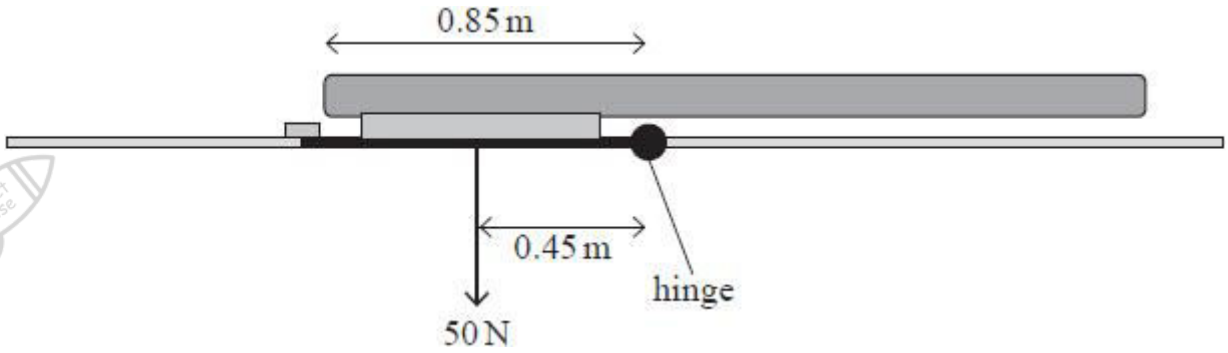
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(b) The weight of the board and rail is 50 N. The weight acts at a distance of 0.45 m from the hinge as shown.

The total length of the ladder is 2.70 m, with 0.85 m to the left of the hinge.

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The weight of the ladder is 54 N. Assume the ladder is uniform.

(i) Explain why the board and ladder remain in the position shown.

(5)

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(ii) A downward force is required to rotate the board away from the block. This force is applied 0.80 m from the hinge.

Determine the magnitude of this force.

(2)

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Magnitude of force = ...

(Total for question = 6 marks)

Q19.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross

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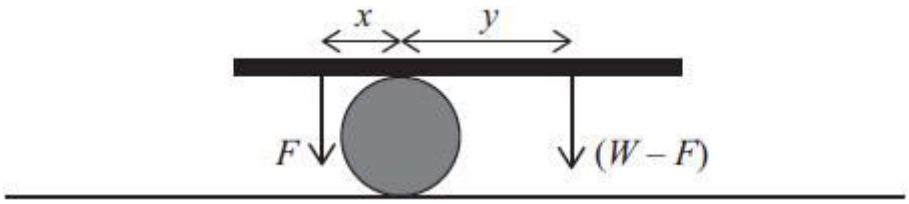
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(☒).

A student puts a piece of wood on top of a cylinder. He stands on the wood with his feet at distances *x* and *y* from the cylinder.

The weight of the student is *W*.

The diagram shows the force of each foot acting on the wood.



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The wood is horizontal and in equilibrium.

The weight of the wood is negligible.

Which of the following expressions is equal to *W*?

☐ **A** $\frac{Fx}{y} + F$

☐ **B** $\frac{Fx}{y} - F$

☐ **C** $\frac{Fy}{x} + F$

☐ **D** $\frac{Fy}{x} - F$

(Total for question = 1 mark)

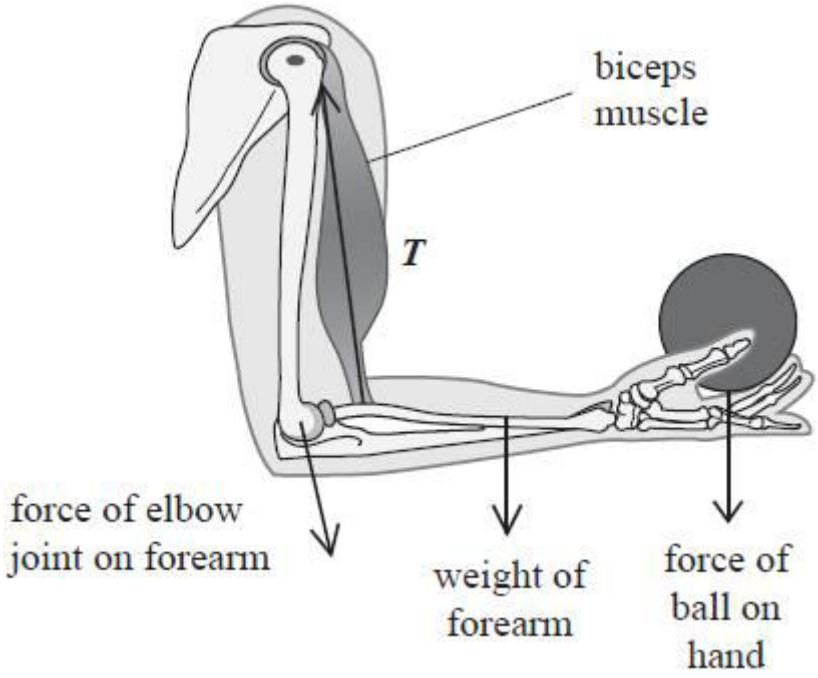
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Q20.

Muscles move body parts by contracting and relaxing. For the forearm to hold a ball in the position shown, the biceps muscle contracts, creating a tension *T* in the muscle as shown.

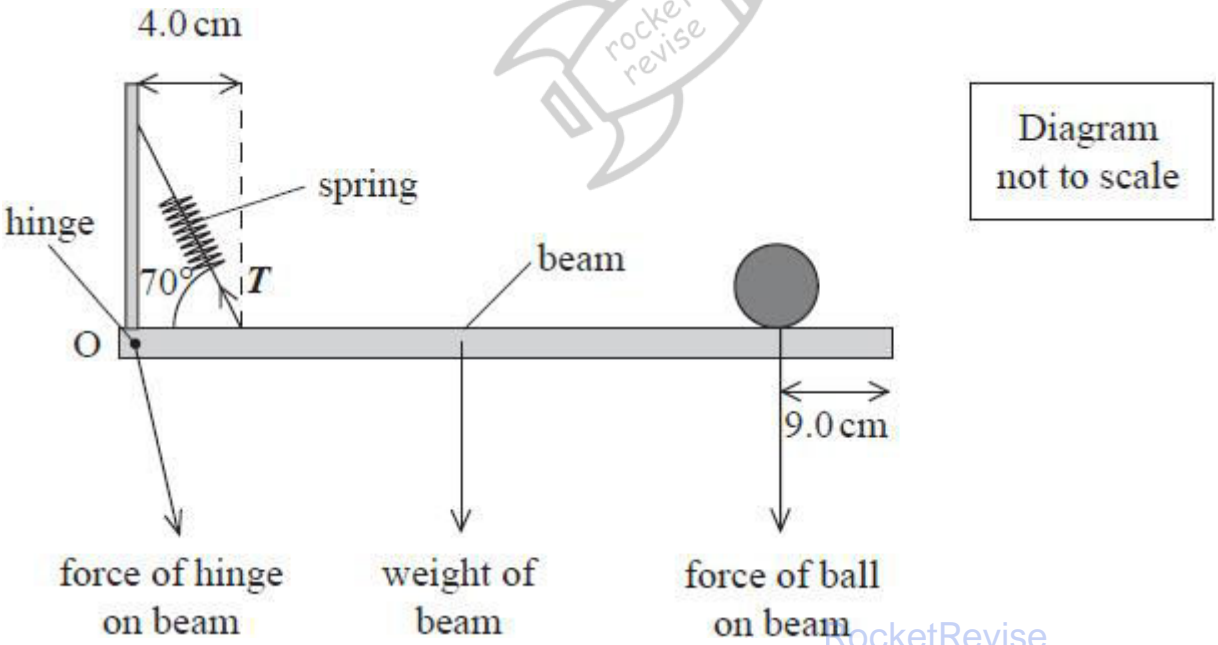
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A student modelled the forces on the forearm using a uniform beam and spring arrangement as shown below. The length and weight of the beam were the same as the length and weight of the forearm.



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(a) It can be assumed that the biceps muscle acts as a spring at an angle of 70° to the beam, 4.0 cm from the pivot O.
Determine the magnitude of *T*. You will need to estimate the total length of the forearm and hand.
force of ball on beam = 4.5 N
weight of beam = 15 N

(5)

Estimate of total length of forearm and hand = ...

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T = ...

(b) Explain a limitation of using a beam to model the forearm.

(2)

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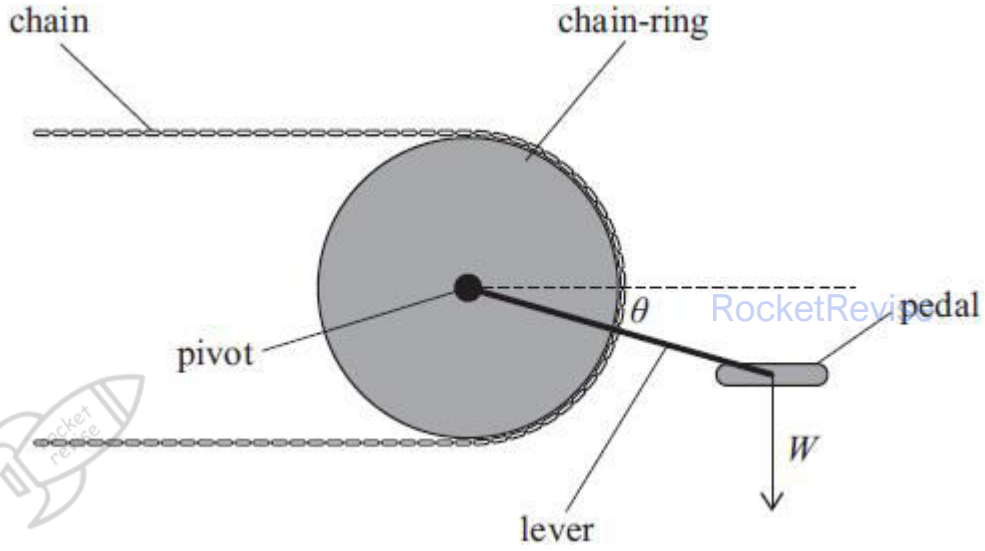
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(Total for question = 7 marks)

Q21.

A bicycle pedal is connected to a circular 'chain-ring' by a lever. A person of weight *W* stands on the pedal.

The lever is at an angle *θ* to the horizontal, as shown.



(a) Determine the moment of *W* about the pivot when *θ* is 24°.

W = 580 N
length of lever = 0.21 m

(3)

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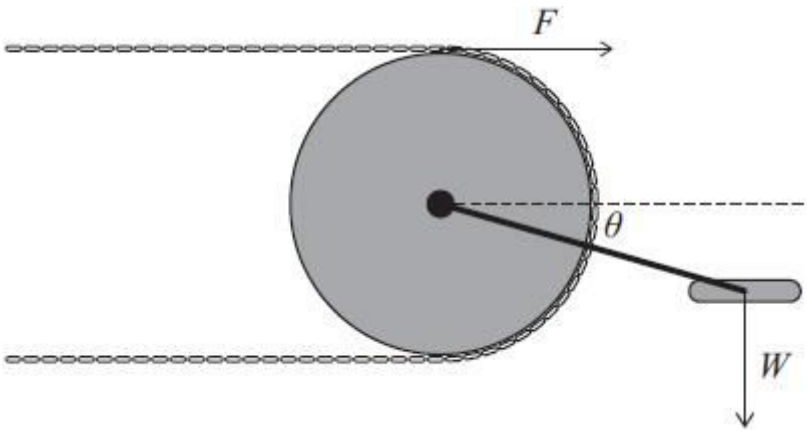
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Moment = ...

(b) The chain-ring exerts a force *F* on the chain, as shown.



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F is proportional to the moment of *W* about the pivot.

Explain how *F* changes as *θ* increases from 0° to 90°.

You do not need to do any calculations.

(2)

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(Total for question = 5 marks)

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Q22.

A student stores her bicycle on a rack.

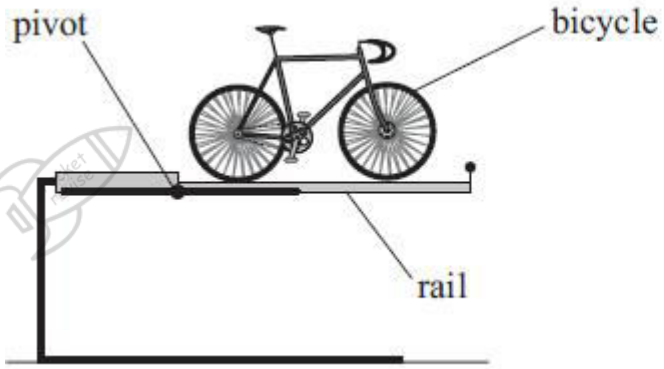
The bicycle rests on a rail that can rotate about a pivot.
To take the bicycle off the rack, the student uses a handle to lower one end of the rail to the ground.

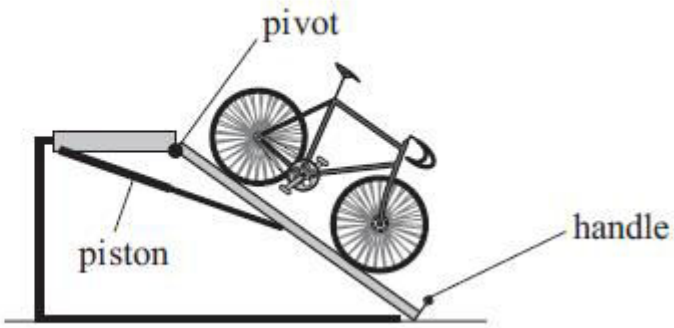
A piston applies a force to the rail as the rail is lowered to the ground.

The diagrams show the bicycle on the rack before and after the rail has been lowered to the ground.

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(a) When the rail is in equilibrium, the principle of moments applies.

State what is meant by the principle of moments.

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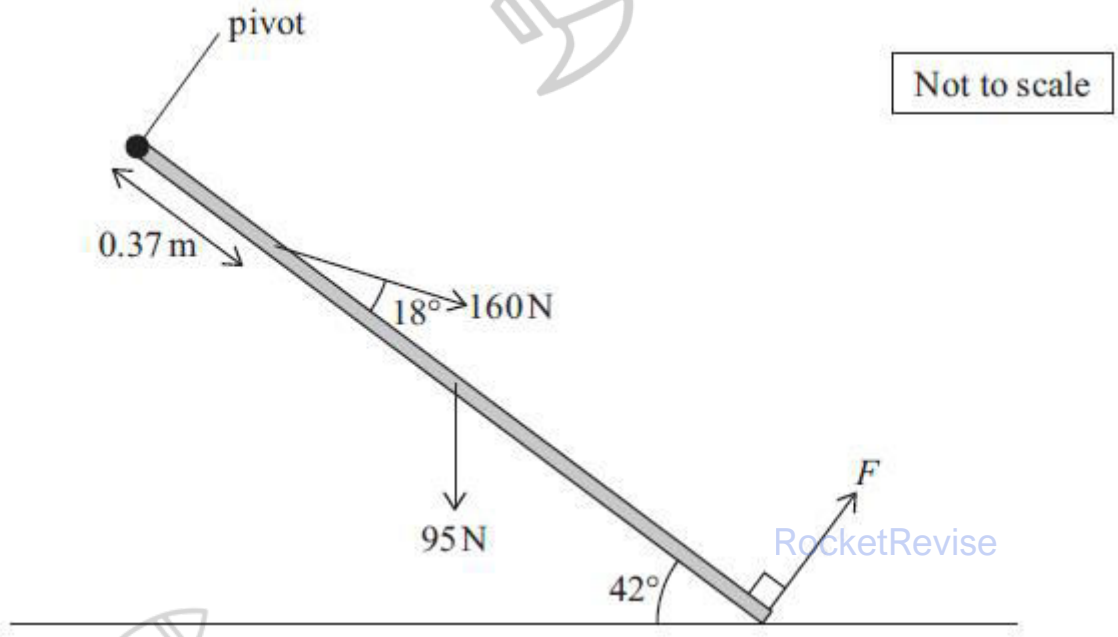
(1)

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(b) After taking the bicycle off the rack, the student lifts the rail back to its original position.

(i) The uniform rail has a length of 1.40 m and a weight of 95 N.
The student applies a force *F* to the handle, perpendicular to the rail. The piston applies a force of 160 N to the rail.
The diagram shows these forces acting on the rail.



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Determine the minimum value of *F* needed to lift the end of the rail off the ground.

(5)

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Minimum value of F = ...

(ii) The student could have lifted the rail by applying a vertical upwards force to the handle.
The force needed to lift the rail is different when the force is vertical compared with when the force is perpendicular to the rail.
Explain how the minimum vertical force needed is different.
You do not need to include any further calculations.

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(3)

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(Total for question = 9 marks)

Q23.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

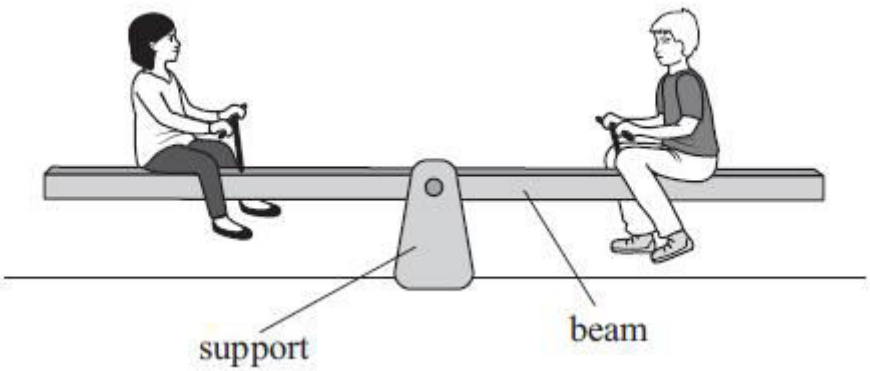
A see-saw is a uniform beam that balances on a support. The beam can rotate around the top of the support.

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Two children of equal weight sit on a see-saw at equal distances from the support, as shown.

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Which row of the table describes the resultant force and resultant moment on the beam?

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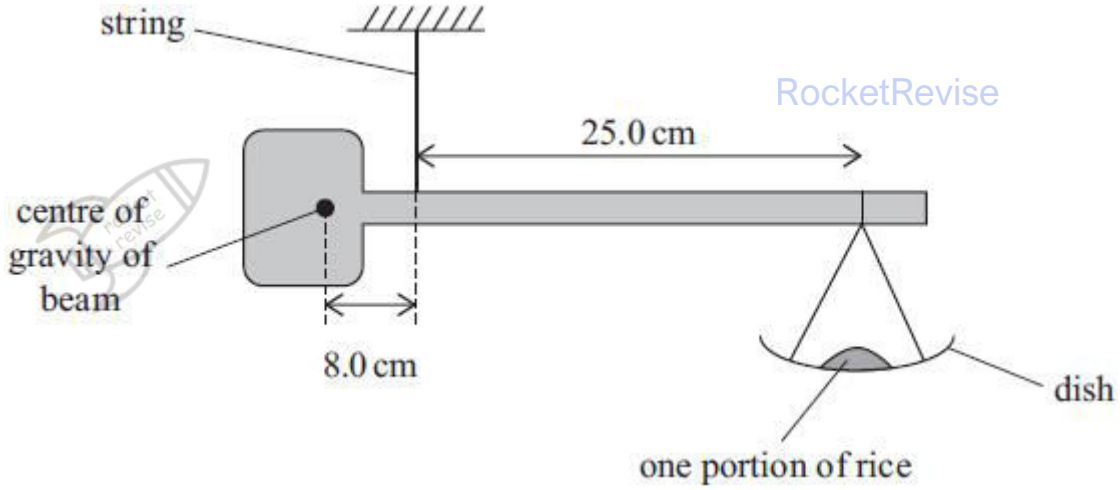
		Resultant force	Resultant moment
☐	A	zero	zero
☐	B	zero	anticlockwise
☐	C	downwards	zero
☐	D	downwards	anticlockwise

(Total for question = 1 mark)

Q24.

A balance is formed by a beam suspended from a string. A dish is hung from the beam and the balance is used to measure a fixed portion of rice.

When the dish contains one portion of rice, the balance is in equilibrium, as shown.



(a) State the conditions needed for an object to be in equilibrium.

(2)

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(b) The dish is moved along the beam so that the horizontal distance from the string to the dish is 7.0 cm.

Rice is added to the dish until the balance is in equilibrium.
Deduce whether the dish now contains 3 portions of rice.
You should include a calculation of the weight of 1 portion of rice.
weight of beam = 3.1 N
weight of empty dish = 0.64 N

(6)

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(Total for question = 8 marks)

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